

00_WM.R

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Thu Mar 23 11:41:10 2017

```
### Willkommen in R!

# Clean previous variables
rm(list=ls(all=TRUE))

# STEP 1 Use R as handheld calculator, data machine ...

#basic arithmetics
5+4

## [1] 9

sqrt(9)

## [1] 3

exp(1)

## [1] 2.718282

log(2.718282)

## [1] 1

x=2
x^2

## [1] 4

#Set up your own data
#Konstruktion eines Spaltenvektors und Zugriff-----
#note that i use = instead of <- cause it is quicker (but dirty)
ALTER=c(21,24,28) #c = combine ...
ALTER=c(22,ALTER,39) #nice trick, to build vector stepwise
ALTER[1:4] #how to select elements

## [1] 22 21 24 28

ALTER[-5]

## [1] 22 21 24 28

ALTER[c(2,5)]

## [1] 21 39
```

```

ALTER[-c(1,3,4)]

## [1] 21 39

GEWICHT=c(56,63,80,49,75)
GRÖSSE=c(1.64,1.73,1.85,1.6,1.81)

#Konstruktion einer Matrix und Zugriff-----
X=cbind(ALTER,GEWICHT,GRÖSSE) #column bind
X[2,3] # how to select again, row/col index

## GRÖSSE
## 1.73

colnames(X)

## [1] "ALTER" "GEWICHT" "GRÖSSE"

namen=c("Gerda","Ka","Ha","Do","Lu")
rownames(X)=namen
X["Ka",] #how to select, new way ..., don't forget the ,

## ALTER GEWICHT GRÖSSE
## 21.00 63.00 1.73

X[c("Ka","Lu"),]

## ALTER GEWICHT GRÖSSE
## Ka 21 63 1.73
## Lu 39 75 1.81

#Konstruktion eines Data Frames-----
SEX = c("w","w","m","w","m")
GEWICHTSDATEN = data.frame(SEX,X)#works like cbind but does not unify data types!
GEWICHTSDATEN

## SEX ALTER GEWICHT GRÖSSE
## Gerda w 22 56 1.64
## Ka w 21 63 1.73
## Ha m 24 80 1.85
## Do w 28 49 1.60
## Lu m 39 75 1.81

str(GEWICHTSDATEN)

## 'data.frame': 5 obs. of 4 variables:
## $ SEX : Factor w/ 2 levels "m","w": 2 2 1 2 1
## $ ALTER : num 22 21 24 28 39
## $ GEWICHT: num 56 63 80 49 75
## $ GRÖSSE : num 1.64 1.73 1.85 1.6 1.81

# STEP 2 - towards the power of charts

```

#Load a package, which has to be installed first! -> Pakete | Installiere Paket ...

```
library(mosaic)
```

```
## Loading required package: dplyr
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##   filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##   intersect, setdiff, setequal, union
```

```
## Loading required package: lattice
```

```
## Loading required package: ggplot2
```

```
## Loading required package: mosaicData
```

```
## Loading required package: Matrix
```

```
##
```

```
## The 'mosaic' package masks several functions from core packages in order  
## to add additional features.
```

```
## The original behavior of these functions should not be affected by this.
```

```
##
```

```
## Attaching package: 'mosaic'
```

```
## The following object is masked from 'package:Matrix':
```

```
##
```

```
##   mean
```

```
## The following objects are masked from 'package:dplyr':
```

```
##
```

```
##   count, do, tally
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##   binom.test, cor, cov, D, fivenum, IQR, median, prop.test,
```

```
##   quantile, sd, t.test, var
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##   max, mean, min, prod, range, sample, sum
```

```
library(corrplot)
```

```
# Load data
```

```
tips = read.table("tips.csv", header=TRUE, sep=",", na.strings="NA", dec=".",
strip.white=TRUE)
head(tips)
```

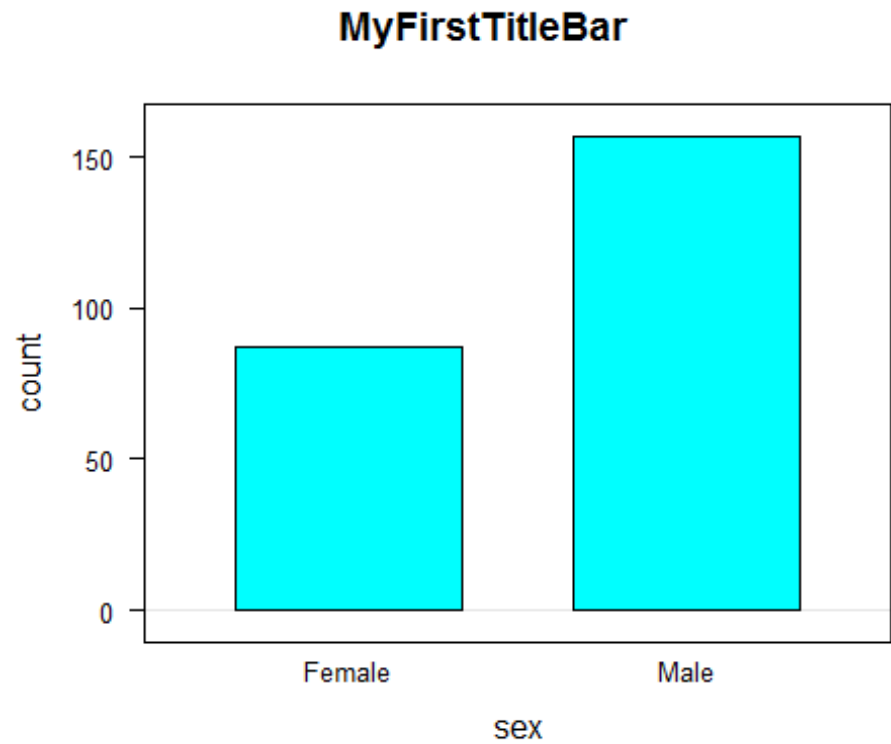
```
##   X total_bill  tip    sex smoker day   time size
## 1 1      16.99 1.01 Female    No  Sun  Dinner    2
## 2 2      10.34 1.66   Male    No  Sun  Dinner    3
## 3 3      21.01 3.50   Male    No  Sun  Dinner    3
## 4 4      23.68 3.31   Male    No  Sun  Dinner    2
## 5 5      24.59 3.61 Female    No  Sun  Dinner    4
## 6 6      25.29 4.71   Male    No  Sun  Dinner    4
```

```
summary(tips)
```

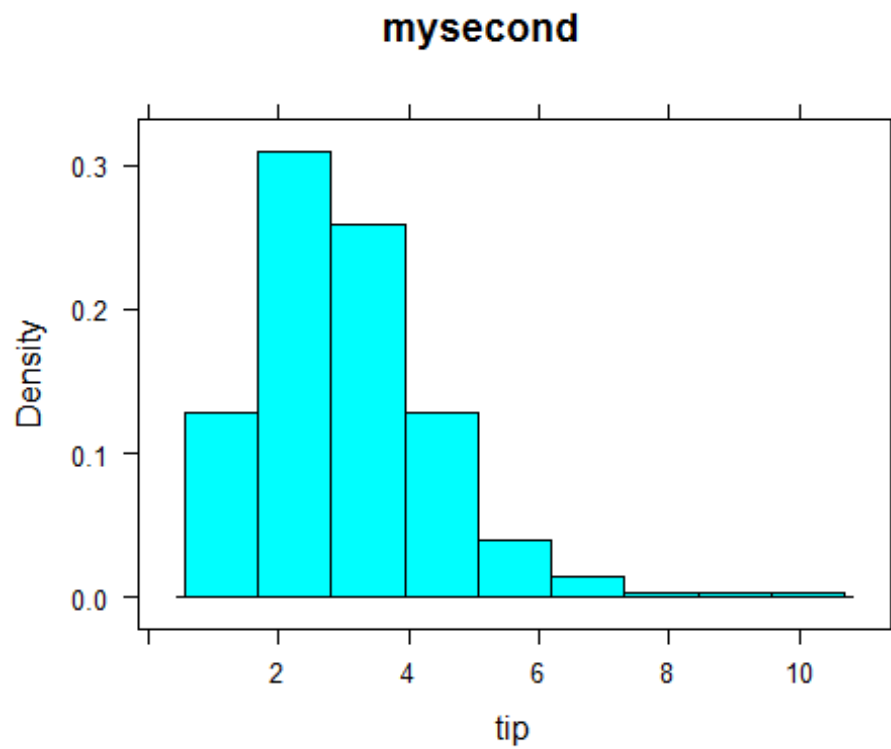
```
##           X           total_bill           tip           sex           smoker
##  Min.      : 1.00      Min.      : 3.07      Min.      : 1.000      Female: 87      No :151
## 1st Qu.: 61.75      1st Qu.:13.35      1st Qu.: 2.000      Male  :157      Yes: 93
## Median :122.50      Median :17.80      Median : 2.900
## Mean    :122.50      Mean    :19.79      Mean    : 2.998
## 3rd Qu.:183.25      3rd Qu.:24.13      3rd Qu.: 3.562
## Max.    :244.00      Max.    :50.81      Max.    :10.000
##   day           time           size
## Fri :19      Dinner:176      Min.    :1.00
## Sat :87      Lunch : 68      1st Qu.:2.00
## Sun :76                               Median :2.00
## Thur:62                               Mean    :2.57
##                               3rd Qu.:3.00
##                               Max.    :6.00
```

STEP 3 - see the power of charts

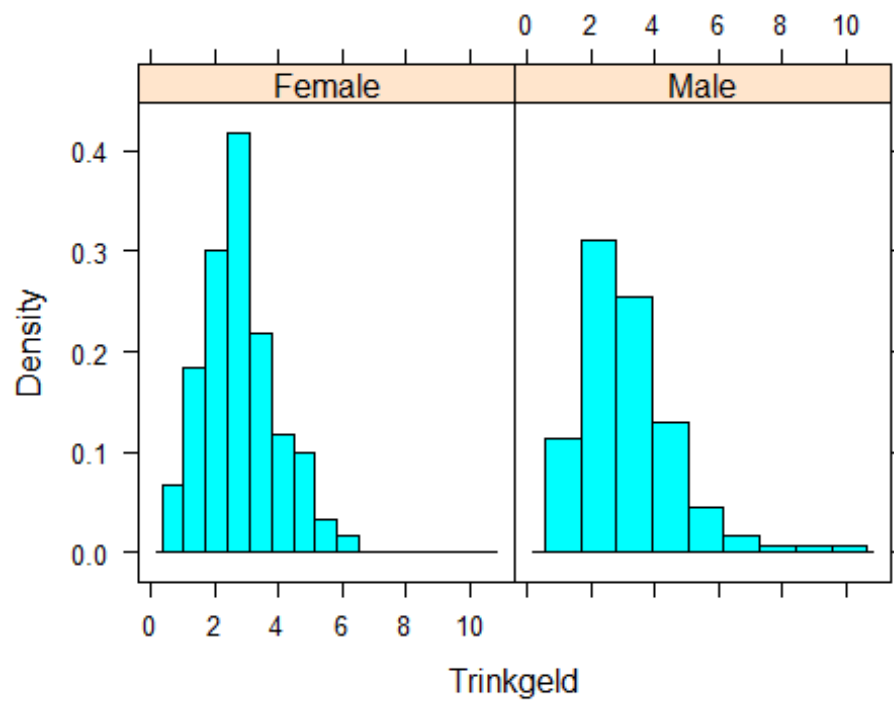
```
bargraph(~sex, data=tips, main = "MyFirstTitleBar")
```



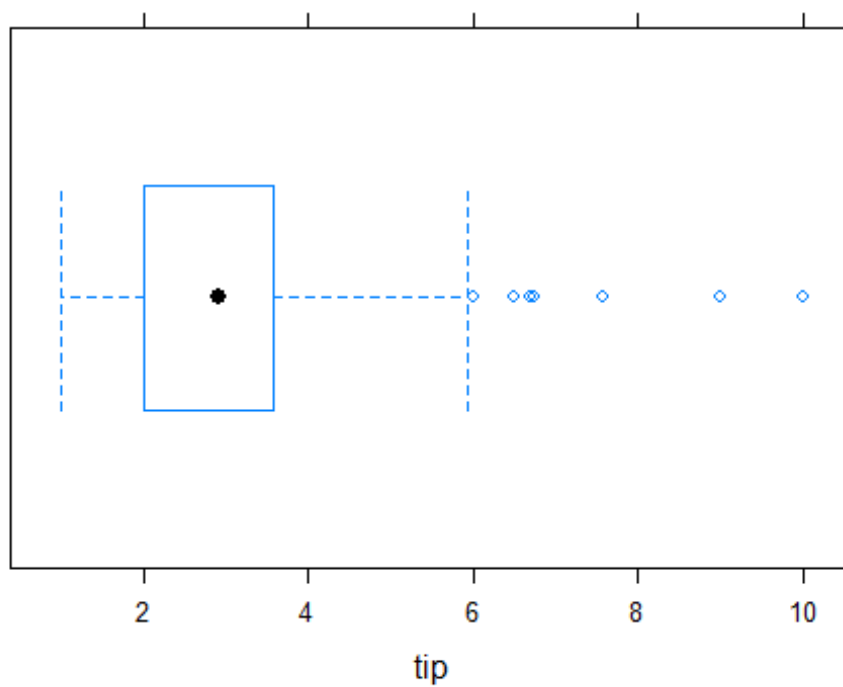
```
x11()  
histogram(~tip, data=tips, main = "mysecond")
```



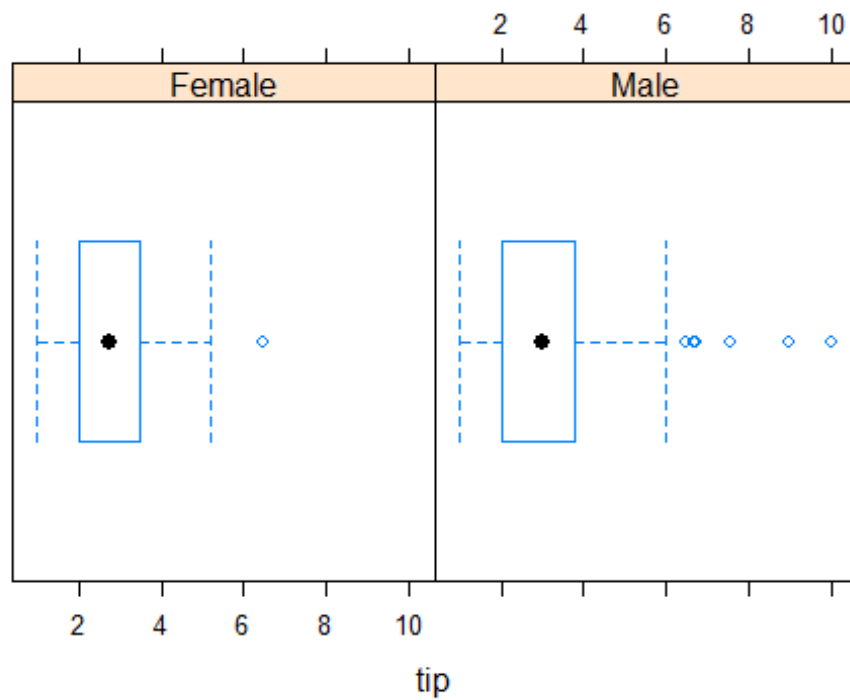
```
x11()  
histogram(~tip | sex, data=tips, xlab = "Trinkgeld")
```



```
x11()  
#Boxplot mit metrischen Daten  
bwplot(~tip, data=tips)
```

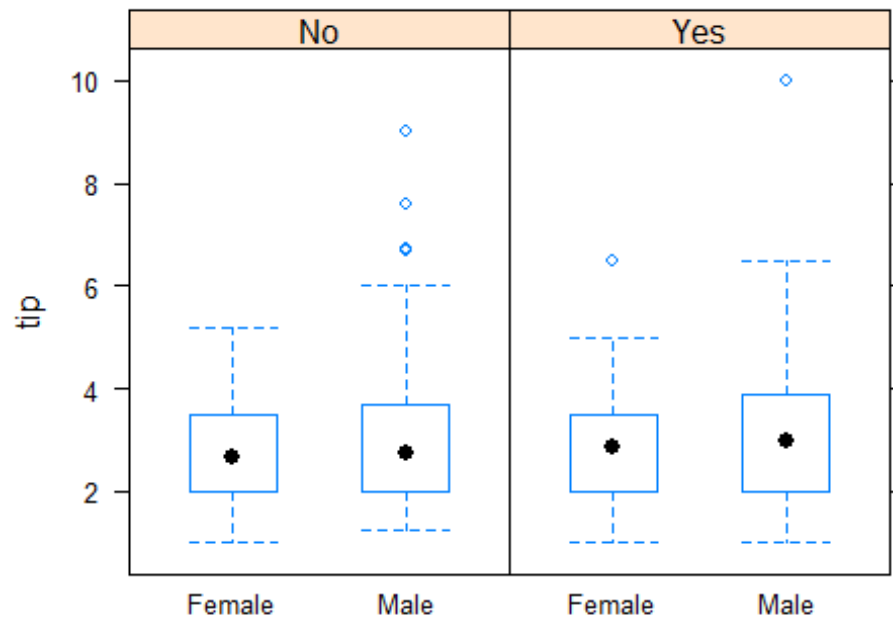


```
x11()
#Boxplot mit metrischen Daten für Gruppen
bwplot(~tip | sex, data=tips)
```

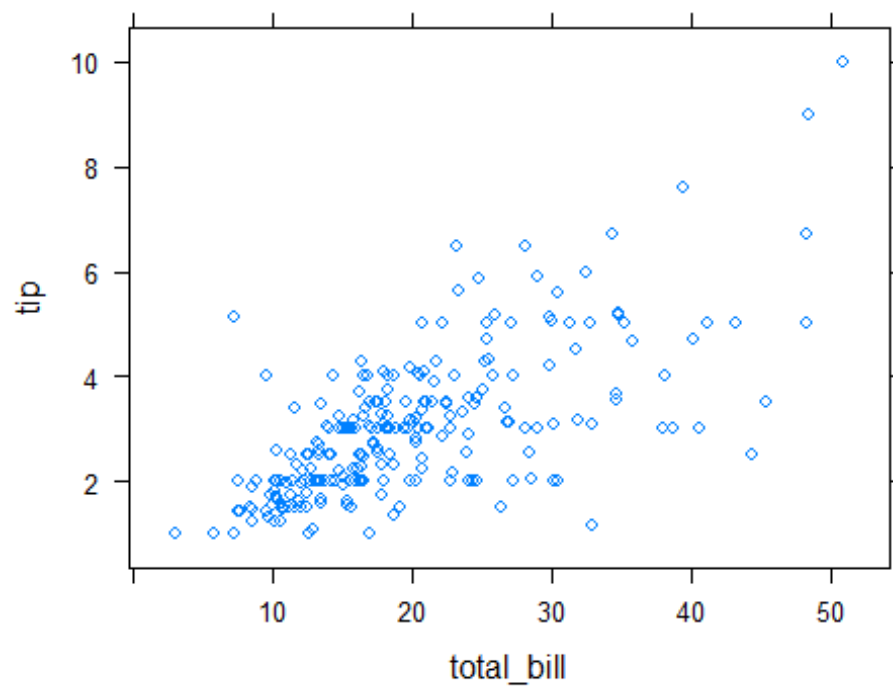


```
x11()
bwplot(tip~sex | smoker, data=tips, main = "Ja ja die Raucher")
```

Ja ja die Raucher



```
x11()  
#Scatterplot (Streudiagramm) mit zwei metrischen Variablen  
xyplot(tip~total_bill, data=tips)
```

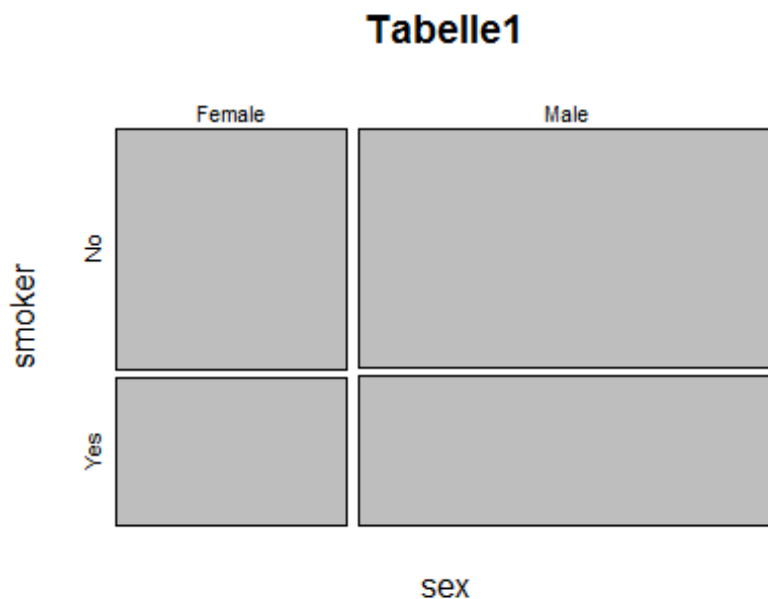


```
x11()

#mosaicplot mit zwei kategorialen Variablen
#Vorher muss eine Tabelle mit dem Befehl tally generiert werden
tally(sex~smoker, data=tips)

##           smoker
## sex      No Yes
##  Female  54  33
##   Male   97  60

#Tabelle wird einem Objekt zugewiesen
Tabelle1<-tally(sex~smoker, data=tips)
mosaicplot(Tabelle1)
```



```
#differenzierte stats
favstats(tip~sex, data=tips)

##      sex min Q1 median   Q3  max    mean      sd  n missing
## 1 Female   1  2   2.75  3.50   6.5 2.833448 1.159495  87        0
## 2  Male    1  2   3.00  3.76  10.0 3.089618 1.489102 157        0

x11()
corrplot(cor(tips[,c("total_bill", "tip", "size")]))
```

